Lahore University of Management Sciences
School of Science and Engineering
Department of Artificial Intelligence

Programming Assignment 2
Question 2 Report

Moving MNIST: Motion, Symmetry and Reach

Course	AI651 – Deep Learning for Space, Time and Graphs
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GitHub repository	https://github.com/MuhammadBinTariq/DL4STG-PA2
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1 Setup and Training Motion

Moving MNIST uses a fixed 8,000/1,000/1,000 train/validation/test video split, with twenty 64×64 grayscale frames per video. Five inputs forecast five frames; at $3\times$ and start 1, only two targets exist. All learned models use Adam at 10^{-3}, batch size 32, 30 epochs, final weights and seeds 0/1 on an NVIDIA A100-SXM4-40GB. Loss weights videos equally after averaging their valid targets. Tests use all 1,000 videos at start 1; entries are mean (seed 0, seed 1). $E^{(5)}$ and $E^{(2)}$ average five and two horizons respectively.

$$C_T = \mathrm{MSE}(f(TX), T(f(X))), \qquad E_T = \mathrm{MSE}(f(TX), T(Y)).$$

Table 1. Transformation and window checks. Marker locations are zero-based; window frame numbers are one-based.

Check	Expected	Measured
(a) max $\lvert M(M(x)) - x\rvert$	0	0.0
(b) max $\lvert R^4(x) - x\rvert$	0	0.0
(c) max $\lvert MRM(x) - R^{-1}(x)\rvert$	0	0.0
(d) Bright pixel of $M(p)$	(7, 10, 43)	(7, 10, 43)
(e) Bright pixel of $R(p)$	(7, 43, 10)	(7, 43, 10)
(f) $1\times$ inputs / targets	1, 2, 3, 4, 5 / 6, 7, 8, 9, 10	1, 2, 3, 4, 5 / 6, 7, 8, 9, 10
(g) $2\times$ inputs / targets	1, 3, 5, 7, 9 / 11, 13, 15, 17, 19	1, 3, 5, 7, 9 / 11, 13, 15, 17, 19
(h) $3\times$ inputs / targets	1, 4, 7, 10, 13 / 16, 19	1, 4, 7, 10, 13 / 16, 19

All three group-relation checks passed with exact equality. Optical flow uses the specified Farneback settings on the first 50 training videos.

Q2.1: Training motion statistics

Figure 1(a) shows samples in every direction range, but the bars have different heights. Thus, directions are broadly represented without being equally common. Mirroring or rotation can change their relative frequencies. The transformed motion directions would therefore likely be familiar, although this does not guarantee successful forecasting.

From Figure 1(b), a typical speed is 3-4 pixels/frame, with a median of 3.32. Scaling the unrounded median gives nominal references of 6.64 and 9.95 pixels per observed step at $2\times$ and $3\times$. These are scaled references, not separately measured flow medians.

Where digits overlap, one displacement vector may not represent either digit's motion accurately. This could bias both histograms and the median. The size of this bias was not quantified, so the conclusions remain tentative.

Q2.2: Effects of the transformations

Mirroring changes digit appearance and reverses the horizontal motion component; rotation changes appearance and turns each direction by $90°$. Both preserve speed, so the speed histogram should remain largely unchanged while direction frequencies may change.

Speed-up retains existing frames at larger intervals. It preserves digit shape but increases displacement between observations.

Together, these provide useful controlled comparisons: mirroring/rotation test appearance and orientation changes at unchanged speed, while speed-up tests larger movements at unchanged shape. However, mirroring and rotation do not isolate appearance from motion direction.

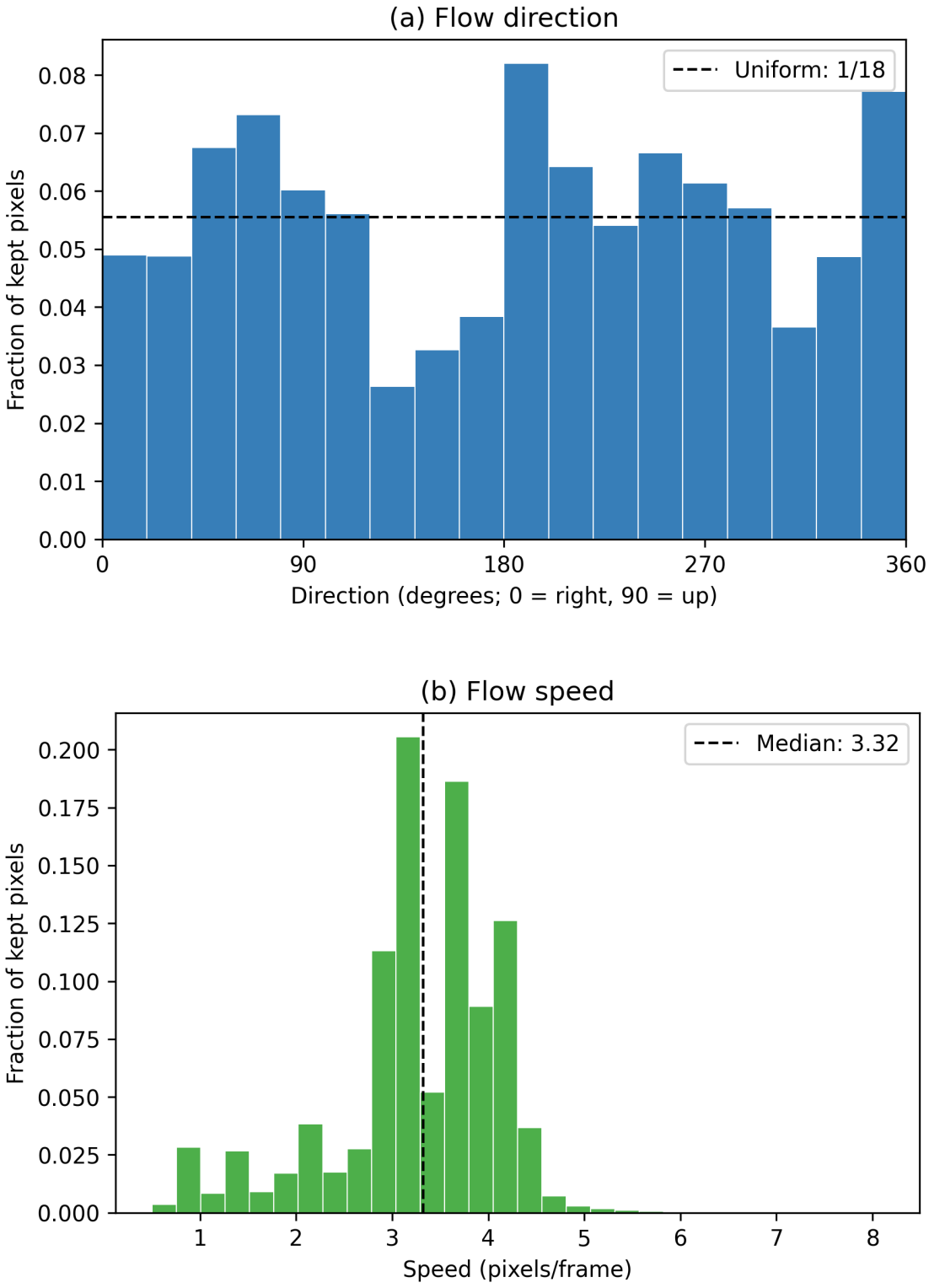


Figure 1. Optical flow from 50 training videos and 950 consecutive frame pairs. The 103,050 retained pixels have intensity above 0.2 at the same location in both frames and flow magnitude above 0.5 pixels/frame. Bars show fractions of retained pixels; median speed is 3.32 pixels/frame. Direction uses 18 bins of $20°$, with $0°$ right and $90°$ up.

2 Baseline ConvLSTM

The baseline has two 64-channel ConvLSTM layers and the prescribed nine-step autoregressive rollout without teacher forcing; outputs of updates 5–9 give the five forecasts.

Q2.3: Model A's reach

Pixel unshuffle changes the 64×64 image into a 16×16 grid, placing each 4×4 patch into channels. Therefore, one folded-grid position corresponds to four original pixels.

A 3×3 convolution moves information by at most one folded-grid position along each axis. Paths from the current input or the previous layer-1 hidden state traverse both ConvLSTM layers; the previous layer-2 hidden state traverses only layer 2. Thus, the longest within-step path contains two spatial convolutions.

The reach is therefore $2 \times 4 = 8$ original pixels per axis, or $8\sqrt{2} \approx 11.3$ pixels diagonally. The patch width is not added, and the 1×1 readout and shuffle operations do not increase this reach.

Table 2. Original $1\times$ forecasting errors and trainable parameters. Learned entries are mean (seed 0, seed 1); references are deterministic.

(a) First three horizons

Model	MSE_1	MSE_2	MSE_3
A	0.0170 (0.0178, 0.0163)	0.0207 (0.0212, 0.0201)	0.0230 (0.0234, 0.0226)
Blank	0.0426	0.0426	0.0426
Copy-last	0.0492	0.0604	0.0669

(b) Later horizons and mean error

Model	MSE_4	MSE_5	$E^{(5)}$
A	0.0247 (0.0250, 0.0245)	0.0265 (0.0266, 0.0263)	0.0224 (0.0228, 0.0219)
Blank	0.0425	0.0425	0.0426
Copy-last	0.0708	0.0732	0.0641

(c) Trainable parameters

Model	Parameters
A	480784 (480784, 480784)
Blank	0
Copy-last	0

Q2.4: Baseline performance

Model A outperforms Blank and Copy-last at every horizon. Its mean $E^{(5)}$ is 0.0224, compared with 0.0426 and 0.0641 respectively.

Its error increases from $\text{MSE}_1 = 0.0170$ to $\text{MSE}_5 = 0.0265$. Both seeds show this trend, consistent with accumulating autoregressive error. Blank remains approximately constant, while Copy-last becomes progressively worse.

In Figure 2, overlapping digits become blurry blobs with merging and loss of identity. However, blur is also visible without overlap, so overlap alone does not explain these failures.

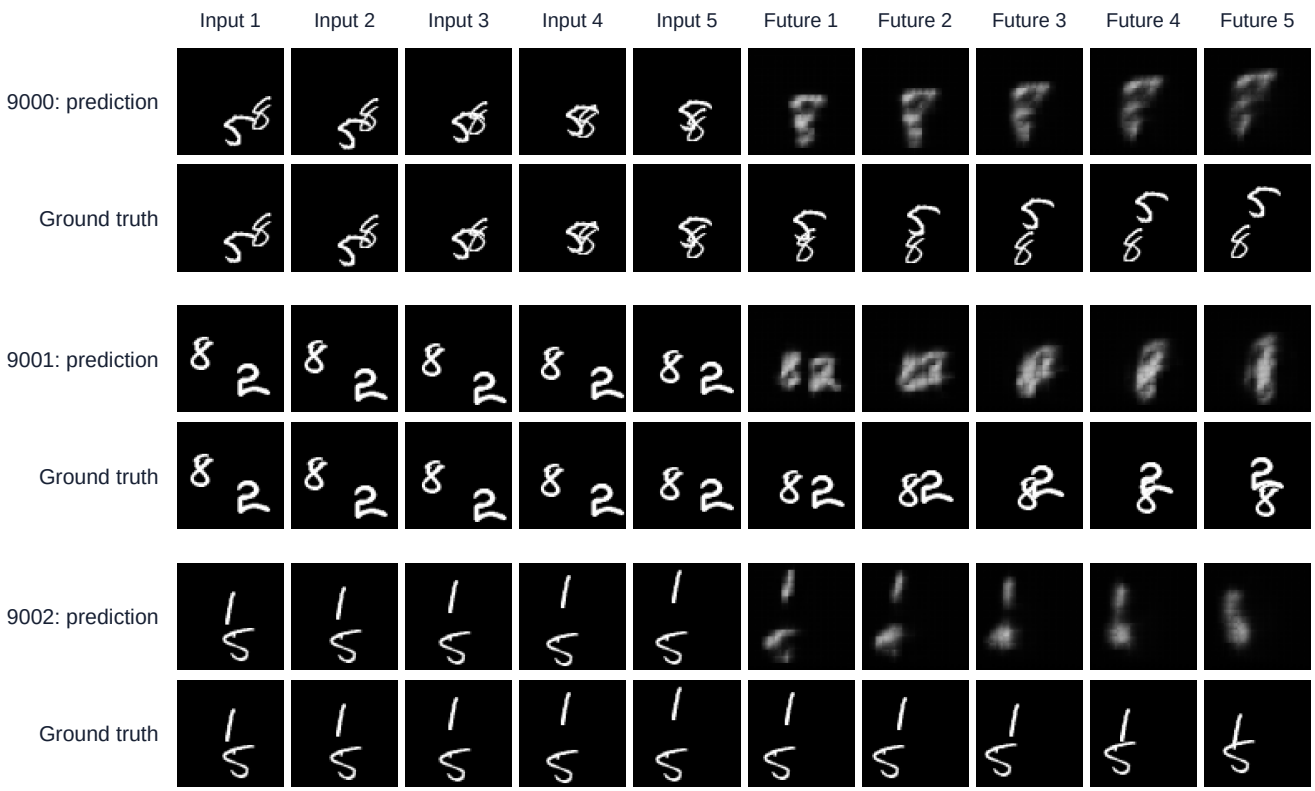


Figure 2. Model A, seed 0, final epoch 30, on original $1\times$ test videos 9000–9002. Columns 1–5 are observed frames and columns 6–10 are future frames. Each pair of rows shows predictions and ground truth. The examples include overlapping digits.

Q2.5: Predictions before testing the baseline

According to my understanding of Model A, I predict that it will fail both the mirror and rotation tests. Although Figure 1 shows broad coverage of optical flow directions, the architecture does not explicitly enforce consistency under mirroring or rotation. Thus, broad direction coverage alone may not be sufficient for consistent predictions on transformed videos.

For speed transformations, I expect noticeable degradation at 2x and the clearest breakdown at 3x. Typical training speeds are around 3-4 pixels/frame, whereas the scaled median references are approximately 6.64 and 9.95 pixels/frame. The 2x reference is within the model's eight-pixel reach along each axis, but these larger displacements are less familiar from training. At 3x, near-axis-aligned movement can exceed that reach, although some diagonal movements remain within the spatial support. Therefore, unfamiliar speeds and limited reach may both contribute to increased blur, misplaced predictions and loss of digit identity

Prediction preserved verbatim from pre-test commit `31adeaa`.

3 Testing the Baseline

Table 3. Original and transformed errors, with mirror/rotation consistency gaps at $1\times$.

(a) Forecasting errors

Model	$E^{(5)}$	E_M	E_R
A	0.0224 (0.0228, 0.0219)	0.0230 (0.0234, 0.0226)	0.0239 (0.0244, 0.0234)
Blank	0.0426	0.0426	0.0426
Copy-last	0.0641	0.0641	0.0641

(b) Consistency gaps

Model	C_M	C_R
A	0.00166 (0.00165, 0.00168)	0.00286 (0.00283, 0.00289)
Blank	0.00×10^{0}	0.00×10^{0}
Copy-last	0.00×10^{0}	0.00×10^{0}

Q2.6: Consistency gap and error

Reading and evaluating on the basis of consistency gap alone is problematic because transformation equivariance does not guarantee accurate forecasts. Blank and Copy-last both have $C_T = 0$, yet their transformed errors are 0.0426 and 0.0641, respectively. Thus, E_T needs to be reported alongside C_T to check whether the forecasts also match the ground truth.

Q2.7: Video 9925 observations

In Figure 3(d), video 9925 has a rotation consistency gap of approximately 0.00749. In row 3, where the model receives the rotated inputs, the predicted digits stretch vertically towards each other and partly merge into a blur at later horizons. In row 4, where rotation is applied to the original forecasts, the digits remain separate, although their shapes are still blurry.

Thus, rotation changes the model's prediction of the digits' relative motion and separation. Row 4 preserves separation better, but this does not establish that its motion is correct.

Q2.8: Transformation comparison

Table 3 indicates that rotation is harder for model A: $C_R = 0.00286$, compared with $C_M = 0.00166$, while $E_R = 0.0239$, compared with $E_M = 0.0230$. Both seeds show this trend.

From Q2.1, motion directions are broadly covered but not equally common. From Q2.2, both transformations preserve speed while changing motion direction and digit appearance. Unfamiliar rotated appearances are therefore a plausible contributor, although direction bias may also contribute. These results alone cannot isolate the two effects.

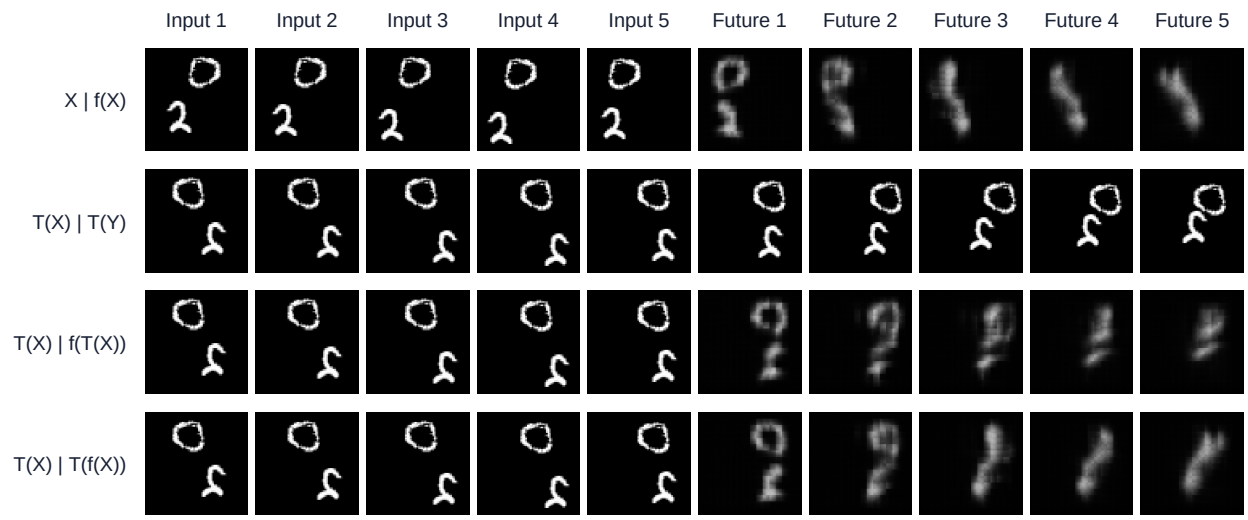


Figure 3(a). Model A, seed 0, $1\times$, video 9257, transformation M. $C_M = 0.00456$, $E_M = 0.0311$. The two largest per-video gaps were selected separately for each transformation.
GIF: outputs/gifs/figure3a_M_video9257.gif.

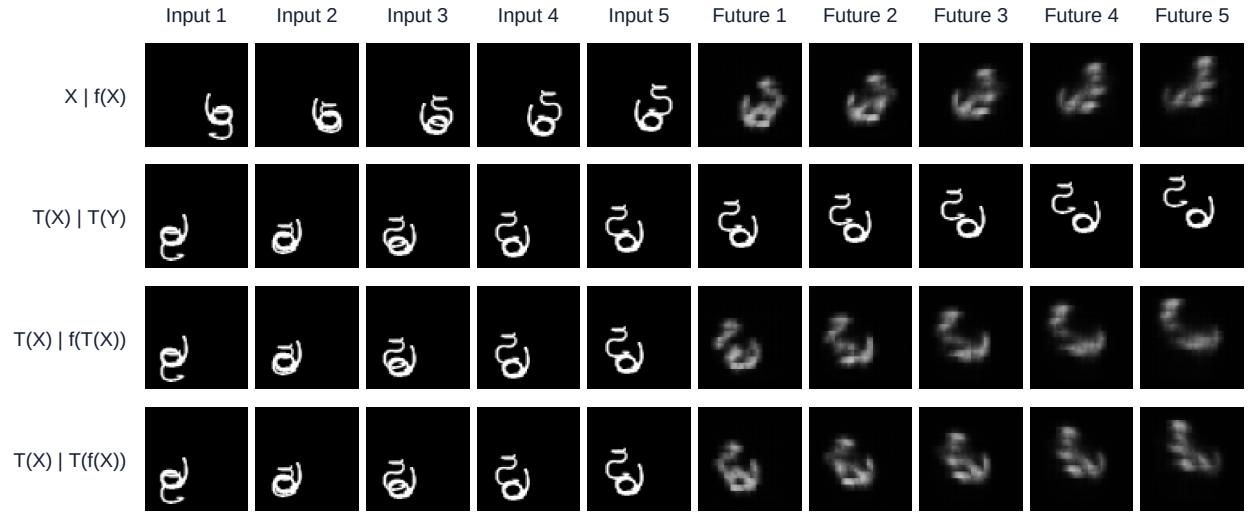


Figure 3(b). Model A, seed 0, $1\times$, video 9348, transformation M. $C_M = 0.00436$, $E_M = 0.0237$. The two largest per-video gaps were selected separately for each transformation.
GIF: outputs/gifs/figure3b_M_video9348.gif.

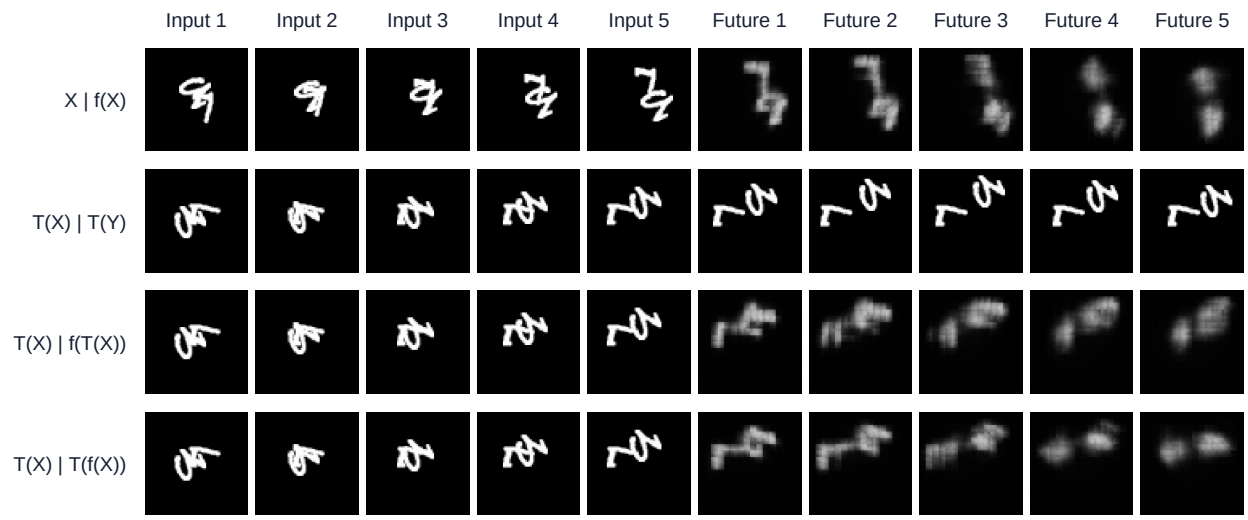


Figure 3(c). Model A, seed 0, $1\times$, video 9497, transformation R. $C_R = 0.00792$, $E_R = 0.0331$. The two largest per-video gaps were selected separately for each transformation.
GIF: outputs/gifs/figure3c_R_video9497.gif.

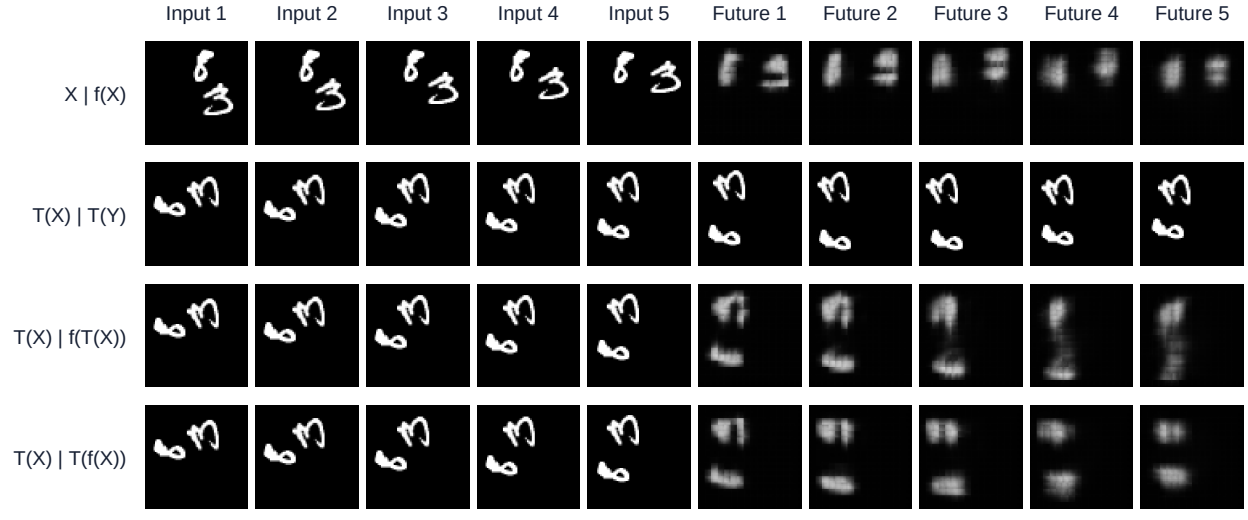


Figure 3(d). Model A, seed 0, $1\times$, video 9925, transformation R. $C_R = 0.00749$, $E_R = 0.0364$. The two largest per-video gaps were selected separately for each transformation.
GIF: outputs/gifs/figure3d_R_video9925.gif.

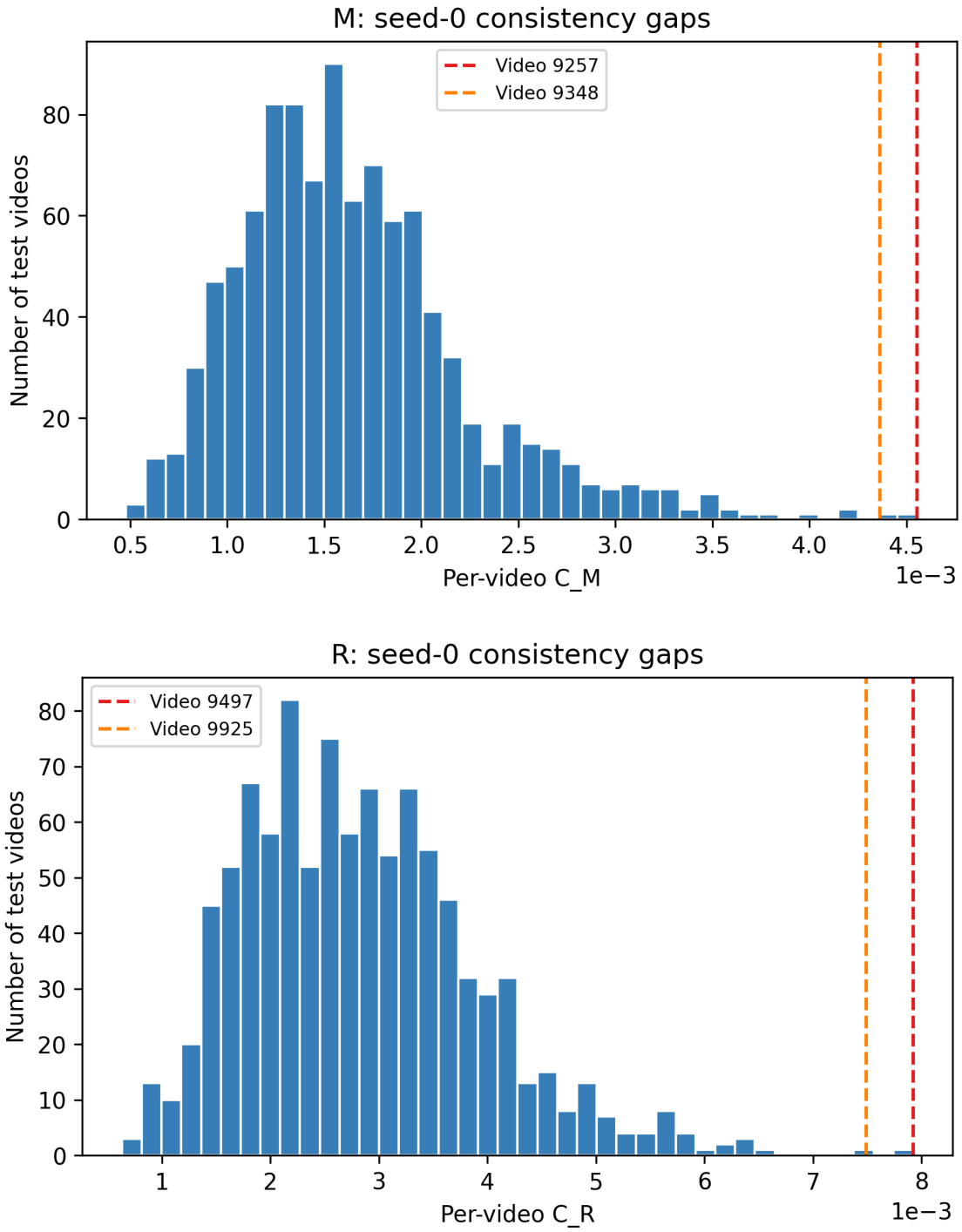


Figure 4. Per-video mirror and rotation consistency gaps for seed-0 model A on all 1,000 original $1\times$ test windows. Lines mark the Figure 3 cases: 9257 and 9348 for M, and 9497 and 9925 for R.

Faster motion

Table 4. Baseline speed tests. Cross-speed comparisons use exactly two forecast horizons.

(a) Two-horizon error

Model	$E^{(2)}, 1\times$	$E^{(2)}, 2\times$	$E^{(2)}, 3\times$
A	0.0189 (0.0195, 0.0182)	0.0443 (0.0436, 0.0451)	0.0503 (0.0499, 0.0506)
Copy-last	0.0548	0.0661	0.0703

(b) Five-horizon error where targets exist

Model	$E^{(5)}, 1\times$	$E^{(5)}, 2\times$
A	0.0224 (0.0228, 0.0219)	0.0482 (0.0475, 0.0489)
Copy-last	0.0641	0.0719

Q2.9: Faster motion and architectural reach

Model A shows substantial degradation at $2\times$ and becomes worse at $3\times$: $E^{(2)}$ increases from 0.0189 to 0.0443 and 0.0503, respectively. Figure 5 already shows blur at $1\times$; at faster speeds, digit shapes deteriorate further. At $3\times$, the predicted 4 changes little between forecasts while its ground-truth position continues changing. This suggests underestimated motion, while the blur suggests uncertainty about shape and position.

The degradation at $2\times$ occurs despite its median speed reference of 6.64 pixels/frame being below the estimated eight-pixel reach per axis. Thus, reach is an upper bound on information movement, not a guarantee of successful forecasting. The estimate does not capture unfamiliar training speeds, and a median does not bound every displacement.

Q2.10: Separating unfamiliar motion from limited reach

These results alone cannot distinguish the two explanations. We could train the same architecture on mixed $1\times$, $2\times$ and $3\times$ windows, then compare it with a larger-reach model trained on the same mixture. Training budgets and evaluation should remain fixed, with comparable parameter counts.

Improvement from speed augmentation would show that unfamiliar motion contributes. Further improvement from the larger-reach model would support architectural reach also being a limitation.

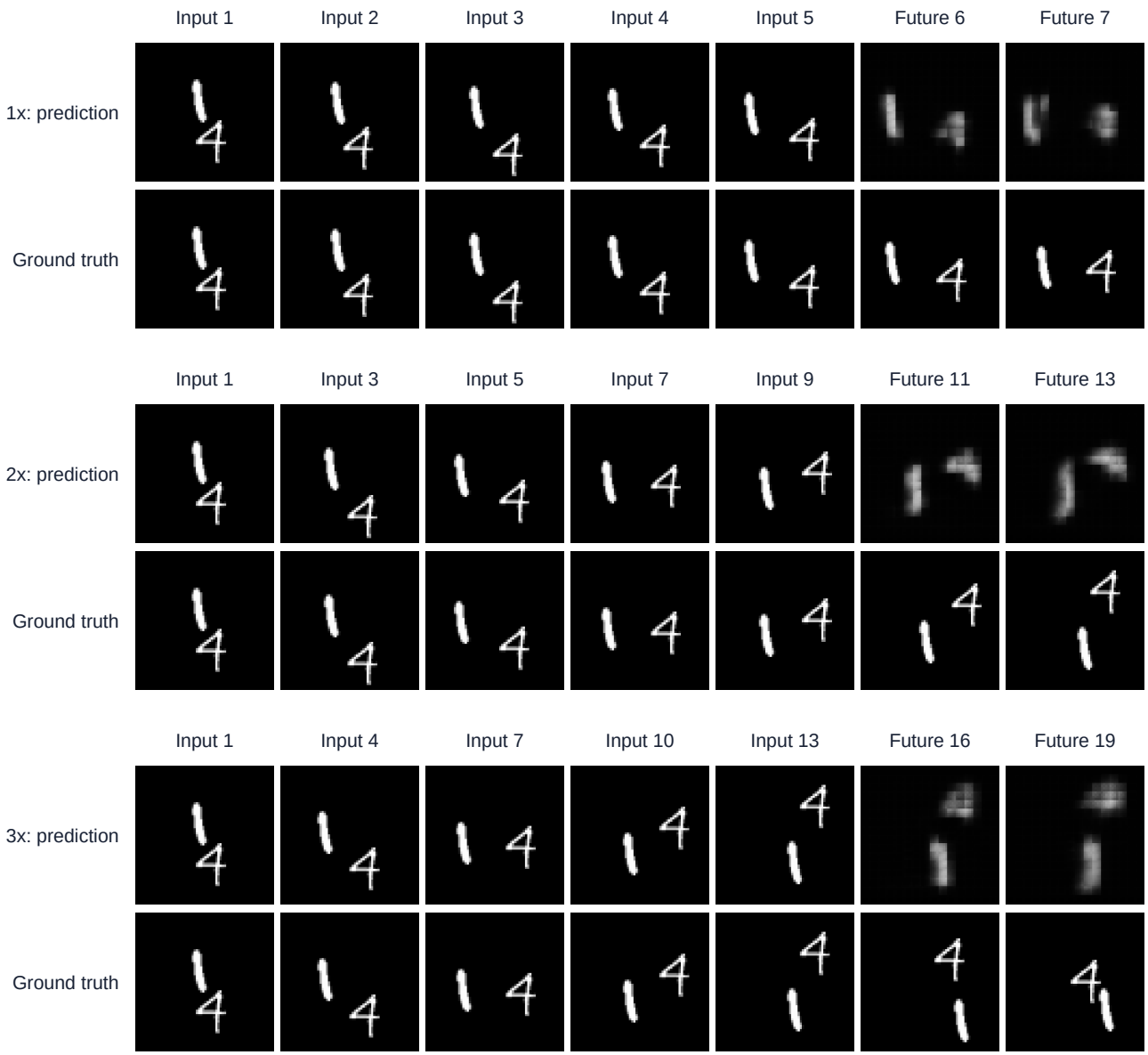


Figure 5. Model A, seed 0, test video 9021. Each speed shows five inputs and the first two predicted/true future frames. Header numbers refer to frames in the original video. The digits do not overlap in any displayed true frame.

4 What Can Data Fix?

Q2.11: Predictions before training B–D

I expect model B to significantly resolve transformation equivariance and reduce the transformation consistency gaps. I expect model C to improve the variable speed forecasts since the same architecture is exposed uniformly to varying motions. Model D's larger reach due to dilation may improve the faster-speed forecasts further.

Prediction preserved verbatim from pre-training commit `2e5b8f5`.

Q2.12: Model D and its reach

Model D keeps the baseline's gate equations and nine-step rollout, but changes the second layer to a 3×3 convolution with dilation 2 and padding 2. Its sampled offsets are $-2, 0, 2$ rather than $-1, 0, 1$, while the folded grid remains 16×16.

Paths from the current input or previous layer-1 hidden state traverse both convolutions, giving $(1 + 2) \times 4 = 12$ pixels per axis. The previous layer-2 hidden state traverses only its dilation-2 convolution, giving 8 pixels. The longest reach is therefore 12 pixels per axis, or $12\sqrt{2} \approx 17.0$ pixels diagonally, strictly larger than A's reach.

Dilation adds no weights. D therefore retains 480,784 parameters, below the 721,176 limit.

B retains A's architecture and $1\times$ training, adding uniform square-symmetry augmentation. C retains A's architecture and samples speeds uniformly from $\{1, 2, 3\}$; D uses the same speed protocol with the larger reach described above. C and D sample starts 1–14, 1–8 and 1–2 at the three speeds respectively.

Table 5. Models A–D: errors, consistency, parameter count and measured cost. Each cell gives the mean above (seed 0, seed 1).

(a) Original/transformed errors and consistency gaps

Model	$E^{(5)}, 1\times$	E_M	E_R	C_M	C_R
A	0.0224 (0.0228, 0.0219)	0.0230 (0.0234, 0.0226)	0.0239 (0.0244, 0.0234)	0.00166 (0.00165, 0.00168)	0.00286 (0.00283, 0.00289)
B	0.0227 (0.0236, 0.0217)	0.0226 (0.0235, 0.0218)	0.0227 (0.0236, 0.0218)	0.00121 (0.00118, 0.00124)	0.00133 (0.00125, 0.00142)
C	0.0270 (0.0275, 0.0264)	0.0272 (0.0277, 0.0267)	0.0280 (0.0285, 0.0275)	8.63×10^{-4} (8.74×10^{-4}, 8.52×10^{-4})	0.00183 (0.00183, 0.00183)
D	0.0272 (0.0271, 0.0274)	0.0274 (0.0273, 0.0275)	0.0284 (0.0283, 0.0284)	9.18×10^{-4} (0.00109, 7.45×10^{-4})	0.00211 (0.00232, 0.00191)

(b) Two-horizon speed errors, parameters and cost

Model	$E^{(2)}, 1\times$	$E^{(2)}, 2\times$	$E^{(2)}, 3\times$	Params	Train s/epoch	Inference s/1,000
A	0.0189 (0.0195, 0.0182)	0.0443 (0.0436, 0.0451)	0.0503 (0.0499, 0.0506)	480784	6.10 (6.11, 6.09)	0.195 (0.187, 0.202)
B	0.0190 (0.0199, 0.0182)	0.0443 (0.0435, 0.0452)	0.0509 (0.0505, 0.0513)	480784	6.54 (6.56, 6.52)	0.188 (0.190, 0.187)
C	0.0231 (0.0238, 0.0225)	0.0291 (0.0292, 0.0290)	0.0333 (0.0335, 0.0332)	480784	6.21 (6.20, 6.22)	0.191 (0.191, 0.191)
D	0.0245 (0.0243, 0.0247)	0.0288 (0.0286, 0.0290)	0.0325 (0.0324, 0.0325)	480784	6.21 (6.21, 6.21)	0.190 (0.192, 0.188)

Q2.13: Model failures

Failures under mirroring and rotation appear partly appearance related: compared with A, B's consistency gaps decrease (C_M from 0.00166 to 0.00121 and C_R from 0.00286 to 0.00133). The transformed errors also decrease: E_M from 0.0230 to 0.0226 and E_R from 0.0239 to 0.0227. However, augmentation also changes direction coverage, so this comparison cannot isolate appearance completely.

The unfamiliar motion failures are the faster speed failures. With the same architecture, C reduces $E^{(2)}$ from 0.0443 to 0.0291 at $2\times$ and from 0.0503 to 0.0333 at $3\times$. Thus, exposure to faster motion substantially improves forecasting.

There is weaker evidence for architectural reach being the main limitation here. D further reduces the $3\times$ error to 0.0325, but this is a modest improvement over C, rather than C failing and D fully resolving the failure. More training data will carry more distributional information thereby improving the overall learned weights but cannot change the reach which is primarily dependent on the spatial connections (i.e. architecture).

Q2.14: Accuracy on original clips

These changes do cost some accuracy on the original $1\times$ clips. B increases mean $E^{(5)}$ slightly, from A's 0.0224 to 0.0227, although the change is mixed across seeds. C and D increase it to 0.0270 and 0.0272, respectively.

For handling varying speeds, this trade-off seems worthwhile because the faster-speed errors decrease substantially. However, A remains preferable if performance on original $1\times$ clips is the only objective.

5 The Final Model

The final model is symmetrised D, the prescribed default. D gives the lowest mean $3\times$ error in Table 5. No optional D+B run or additional training is used. All eight transformed branches run the complete forecaster before inverse transformation and averaging; they share one set of weights. The group is $G = \{R^a, MR^a : a = 0, 1, 2, 3\}$; MR^a rotates first and then mirrors.

$$\bar{f}(X) = \frac{1}{8} \sum_{g \in G} g^{-1} f(gX).$$

For $h \in G$, set $q = gh$. Right multiplication permutes the group and $g^{-1} = hq^{-1}$, so

$$\bar{f}(hX) = \frac{1}{8} \sum_{q \in G} hq^{-1} f(qX) = h\bar{f}(X).$$

A mirror/rotation is a linear pixel permutation, so it commutes with averaging. Thus $C_M = C_R = 0$ in exact arithmetic for any base forecaster. The measured floating-point gaps are retained in Table 6.

Table 6. Symmetrised B and D; symmetrised D is the final model. Gaps are measured floating-point values. Training time belongs to the underlying model. Each cell gives the mean above (seed 0, seed 1).

(a) Original/transformed errors and consistency gaps

Model	$E^{(5)}$, $1\times$	E_M	E_R	C_M	C_R
Sym B	0.0221 (0.0231, 0.0212)	0.0221 (0.0231, 0.0212)	0.0221 (0.0231, 0.0212)	4.29×10^{-17} (3.93×10^{-17}, 4.64×10^{-17})	4.40×10^{-17} (4.04×10^{-17}, 4.76×10^{-17})
Sym D (final)	0.0272 (0.0270, 0.0273)	0.0272 (0.0270, 0.0273)	0.0272 (0.0270, 0.0273)	3.34×10^{-17} (3.65×10^{-17}, 3.04×10^{-17})	3.44×10^{-17} (3.76×10^{-17}, 3.12×10^{-17})

(b) Two-horizon speed errors, parameters and cost

Model	$E^{(2)}$, $1\times$	$E^{(2)}$, $2\times$	$E^{(2)}$, $3\times$	Params	Train s/epoch	Inference s/1,000
Sym B	0.0186 (0.0195, 0.0178)	0.0435 (0.0428, 0.0442)	0.0498 (0.0496, 0.0500)	480784	6.54 (6.56, 6.52)	1.49 (1.48, 1.50)
Sym D (final)	0.0245 (0.0243, 0.0247)	0.0288 (0.0286, 0.0290)	0.0324 (0.0323, 0.0325)	480784	6.21 (6.21, 6.21)	1.49 (1.49, 1.48)

Q2.15: Learned and built-in symmetry

Model B learns approximate consistency through augmentation, but its gaps remain nonzero: $C_M = 0.00121$ and $C_R = 0.00133$. Built-in symmetry is exact mathematically because transforming the input only reorders the eight aligned forecasts. The measured gaps are around 10^{-17}, reflecting floating-point rounding.

Original and transformed errors improved slightly after wrapping. B's mean original $E^{(5)}$ decreased from 0.0227 to 0.0221. D also improved slightly in full precision, although both values round to 0.0272. Inference time increased from approximately 0.19 to 1.49 seconds per 1,000 videos, roughly eight times, with no additional training.

Averaging can hurt accuracy when branches disagree about digit positions and smear an otherwise good forecast. Here, the mean errors decreased despite the visible blur.

Q2.16: Returning to video 9925

In Figure 6(a4), rows 3 and 4 now agree, with $C_R \approx 5.79\times10^{-17}$. However, the predictions are still blurry and indistinguishable blobs, whereas the true digits remain separate. Thus, the final model does not fully get this video right.

Symmetrisation reduces D's error on this clip from 0.0430 to 0.0388, but this remains worse than A's 0.0364 and B's 0.0316. The wrapper fixed the consistency failure without fixing digit identity and tracking. B performs best on this example; the faster-motion improvements from C and D do not produce a better forecast of this particular $1\times$ clip.

Q2.17: Final verdict

The final model has learned some useful motion patterns, but not a reliable general motion rule. Its $E^{(2)}$ at $3\times$ decreases from model A's 0.0503 to 0.0324, at a cost to original $1\times$ accuracy. However, Figure 6 still shows blurry predictions and loss of digit identity. Exact mirror/rotation consistency is guaranteed by the wrapper and therefore does not establish learned motion. Stronger evidence would be sharp forecasts that maintain digit identity and correct trajectories through overlaps and bounces under unseen conditions. Generalisation to speeds beyond the trained $1\times$-$3\times$ range remains uncertain.

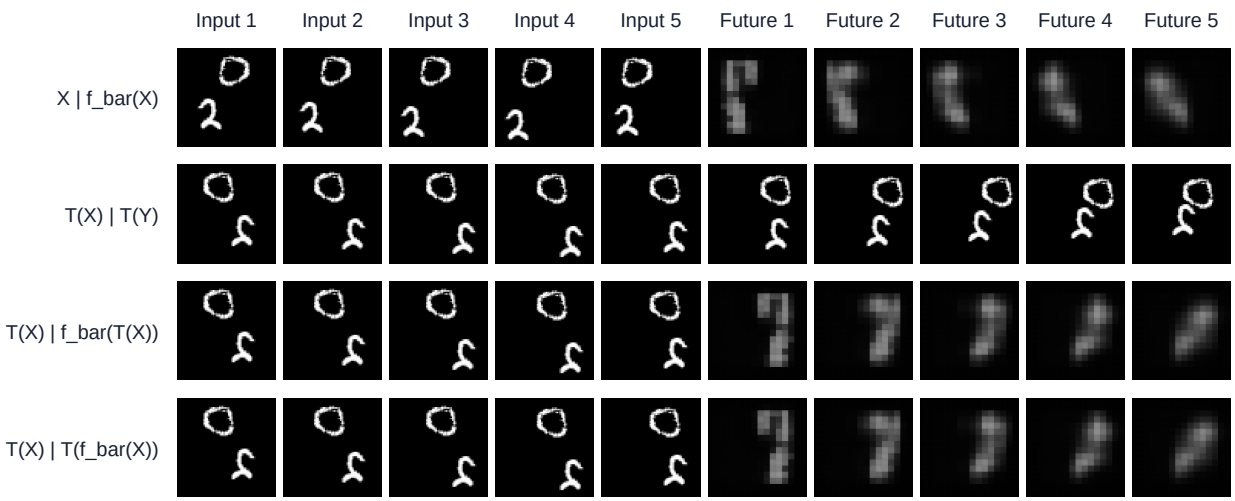


Figure 6(a1). Final model: symmetrised D, seed 0, $1\times$, video 9257, transformation M. $C_M = 4.90\times10^{-17}$, $E_M = 0.0331$. The video is the same baseline Figure 3 case; rows 3 and 4 agree up to floating-point error.
GIF: outputs/gifs/figure6a1_M_video9257.gif.



Figure 6(a2). Final model: symmetrised D, seed 0, $1\times$, video 9348, transformation M. $C_M = 3.84\times10^{-17}$, $E_M = 0.0269$. The video is the same baseline Figure 3 case; rows 3 and 4 agree up to floating-point error.
GIF: outputs/gifs/figure6a2_M_video9348.gif.

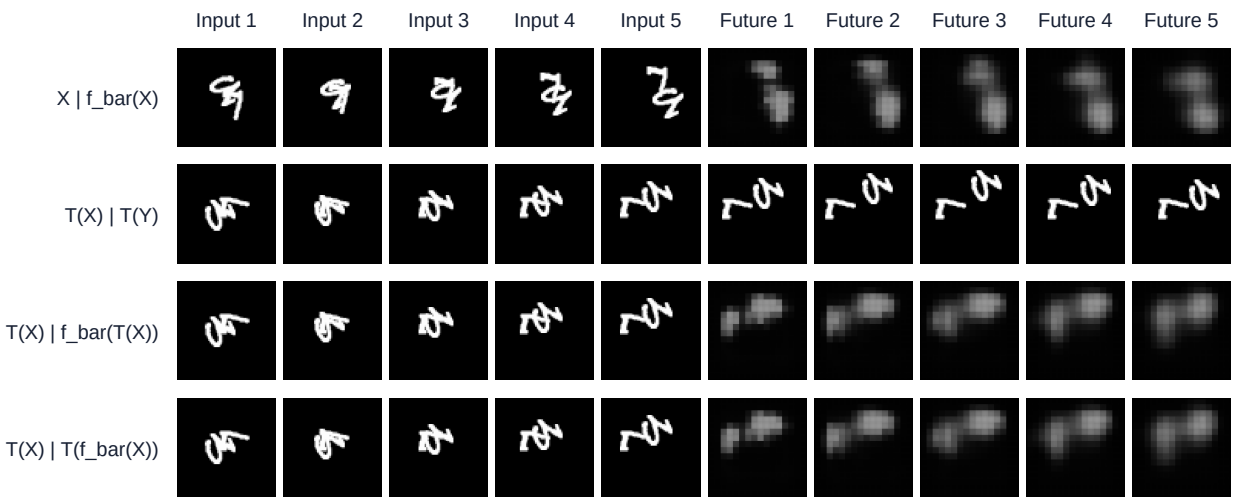


Figure 6(a3). Final model: symmetrised D, seed 0, $1\times$, video 9497, transformation R. $C_R = 5.52\times10^{-17}$, $E_R = 0.0338$. The video is the same baseline Figure 3 case; rows 3 and 4 agree up to floating-point error.
GIF: outputs/gifs/figure6a3_R_video9497.gif.

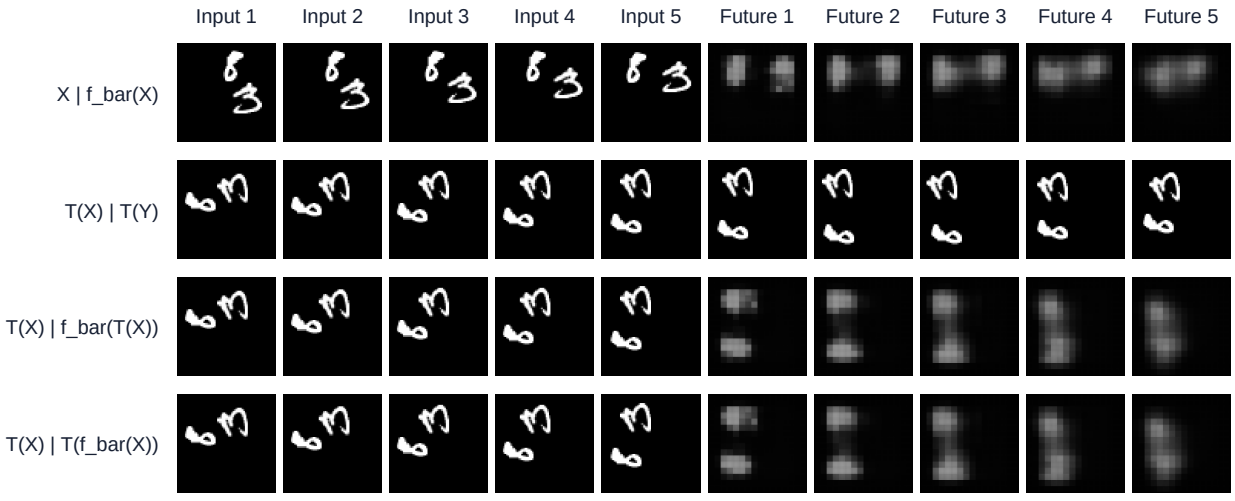


Figure 6(a4). Final model: symmetrised D, seed 0, $1\times$, video 9925, transformation R. $C_R = 5.79\times10^{-17}$, $E_R = 0.0388$. The video is the same baseline Figure 3 case; rows 3 and 4 agree up to floating-point error.
GIF: outputs/gifs/figure6a4_R_video9925.gif.

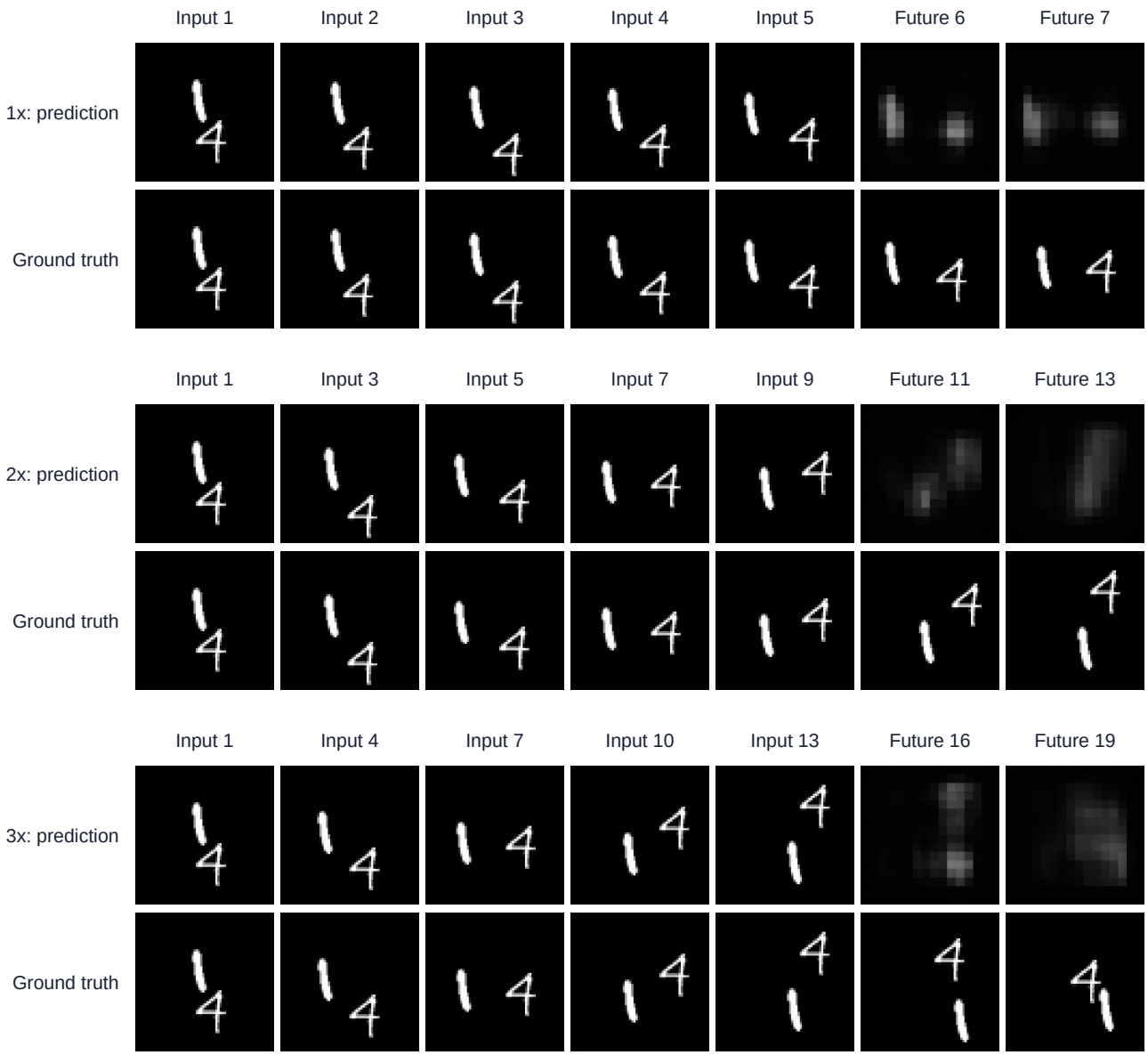
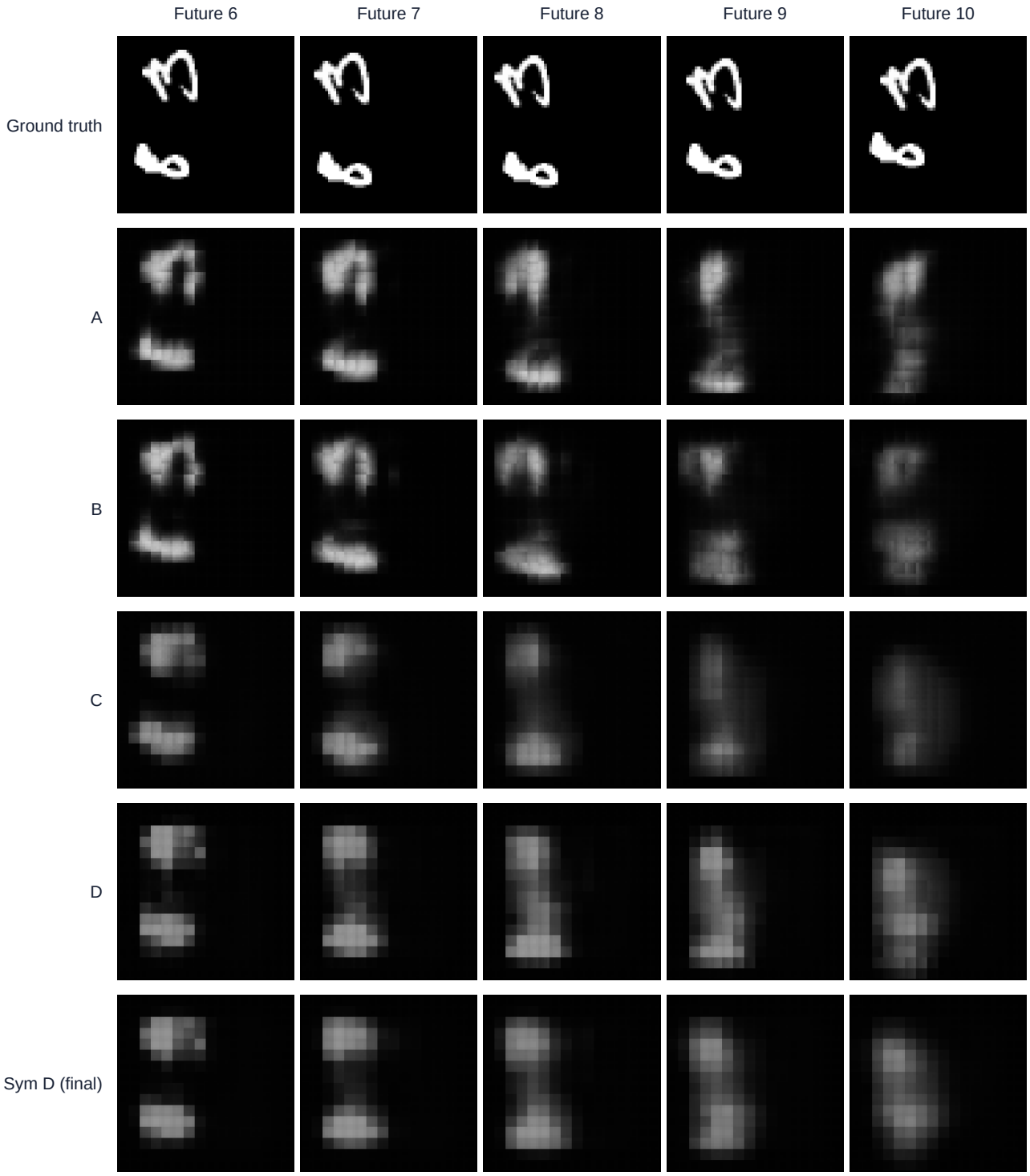


Figure 6(b). Final model: symmetrised D, seed 0, original test video 9021. This repeats Figure 5 on the same video, using the same five inputs and first two future frames at each speed. Header numbers identify original-video frames.

Supplementary comparison for Q2.16



Video 9925 diagnostic. Rotated video 9925, seed 0, future frames 6–10. The ground truth and A/B/C/D/final forecasts show that enforcing consistency does not restore sharp digit identity or guarantee correct motion.